

Notes on Economics

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Chapter 1

Welcome

These notes provide an applied introduction to economics and econometrics, with an emphasis on understanding real-world economic issues and evaluating policy interventions. The material combines economic reasoning with empirical tools commonly used in public policy, health economics, and applied social science.

The focus throughout is on interpretation and judgment rather than formal derivation. Economic models and statistical methods are treated as tools for reasoning about evidence rather than as ends in themselves.

1.1 Who this book is for

These notes are intended for students in economics, public policy, health sciences, and related fields, as well as analysts and researchers working with applied economic data. The material is also suitable for readers with basic quantitative literacy who want to understand how economic evidence is produced and interpreted.

The book does not assume advanced mathematics. Familiarity with introductory statistics is helpful, but not required for all chapters.

1.2 How the book is organized

The chapters progress from concepts to methods to applications.

Early sections introduce economic thinking, scarcity, incentives, institutions, and market structure. Middle sections develop core microeconomic and macroeconomic ideas, including stabilization policy and economic measurement. Later sections focus on empirical methods such as regression, time-series analysis, and quasi-experimental approaches used in policy evaluation.

Although chapters are designed to be readable independently, later material builds on ideas introduced earlier in the book.

1.3 Using figures and visuals

Figures are used throughout the book to support intuition before formal models or code are introduced. Visualizations help clarify relationships between economic actors and variables, illustrate dynamic patterns such as trends and interventions, and support interpretation of statistical results.

Figures are referenced directly in the text and are intended to function as conceptual anchors that readers can revisit while working through later chapters.

1.4 Using R code

Some chapters include R code to demonstrate empirical workflows commonly used in applied economics and policy analysis.

Code chunks are included primarily as examples and templates. Some chunks may use `eval = FALSE` so the book can render without requiring all packages or datasets. Readers who wish to run the code locally can enable evaluation after installing the necessary packages and ensuring data availability.

The emphasis is on understanding the logic of analysis and interpretation rather than reproducing exact numerical outputs.

1.5 Learning goals

By the end of these notes, readers should be able to:

- explain core economic concepts and relate them to real-world policy questions;
- interpret common economic indicators such as GDP, inflation, and unemployment;
- understand the logic and limitations of regression-based analysis;
- recognize issues of trend, autocorrelation, and stationarity in time-series data; and
- assess policy interventions using quasi-experimental reasoning, including interrupted time series analysis.

1.6 A note on interpretation and evidence

Economic and statistical models simplify reality. Throughout the book, emphasis is placed not only on estimation and prediction, but also on interpretation, assumptions, uncertainty, and institutional context.

Economic evidence is always produced within data, measurement, and modeling constraints. Readers are encouraged to approach results critically, paying attention to the assumptions underlying models and the limitations of available evidence.

The goal of the book is not to present economic analysis as a mechanical procedure, but rather as a structured framework for reasoning about complex real-world problems.